Notas de Combinatória 2

Henrique

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0 Informações importantes

Devo guardar notas de aula e exercícios interessantes do curso. Por enquanto nenhuma notação muito esotérica apareceu, se sim devo listá-las aqui. O objetivo dessas notas é serem breves, de forma que é possível recuperar formalmente as provas dos teoremas enunciados. Não serei estritamente rigoroso, o que pode levar a alguns erros. A linguagem pode variar intermintentemente entre português e inglês.

O código fonte desse arquivo pode ser encontrado em https://github.com/hnrq104/comb2/tree/main.

1 Extremal Set Theory

Lemma 1.1. (LYMB) Let $\mathcal{A} \subset \mathcal{P}([n])$ be an anti-chain, then

$$\sum_{A \in \mathcal{A}} \binom{n}{|A|}^{-1} \leqslant 1$$

Proof. (Sketch) Take a random permutation and consider the indicator random variables X_A that the first $|A|$ elements form A. We find that $\mathbb{E} \sum X_A = \sum_{A \in \mathcal{A}} \binom{n}{|A|}^{-1}$, but $\sum X_A \leqslant 1$. $\square$

Corollary 1.2. (Sperner) If $\mathcal{A}$ is an anti-chain, $|\mathcal{A}| \leqslant \binom{n}{n/2}$.

Corollary 1.3. (Littlewood-Offord) Given $a_1, a_2, \dots a_n \in \mathbb{R} \setminus \{0\}$. Consider $x_1, \dots x_n \in \{-1, 1\}$ u.i.i.d r.v's, then $\mathbb{P}(\sum a_i x_i = 0) \leqslant 2^{-n} \binom{n}{n/2}$.

Proof. By simmetry of the x_i, we may assume that all a_i are positive. Now consider the collection $\mathcal{A} := \{A \subset [n] : \sum_{i \in A} a_i - \sum_{i \notin A} a_i = 0\}$. $\mathcal{A}$ is an anti-chain. $\square$

Theorem 1.4. (Erdos-Ko-Rado, Katona) Let $\mathcal{A} \subset \binom{[n]}{k}$ be an intersecting family with $2k < n$, then $|\mathcal{A}| \leqslant \binom{n-1}{k-1}$.

Proof. (Sketch) Take a random circular permutation of $[n]$, let X_A be the indicator of A appearing as an interval on the circle. We find that $\sum X_A \leqslant k$ - partition the intervals that intersect A and use pigeonhole. Calculating expectation and rearranging gives the result. $\square$

Theorem 1.5. (Exercise ERK) There is $n_0(k, l)$ such that if $n \geqslant n_0$ and $\mathcal{A} \subset \binom{[n]}{k}$ is an l-intersecting family, then $|\mathcal{A}| \leqslant \binom{n-l}{k-l}$. Hints:

1. Show that there exists $L \subset [n]$, $|L| = l$ such that $L = A \cap B$ for $A, B \in \mathcal{A}$.
2. Show that either $\mathcal{A} \subset \partial L := \{A \in \binom{[n]}{k} : L \subset A\}$ or $|\mathcal{A}| = O(n^{k-l-1})$.

Proof. For the first step, assume $\mathcal{A}$ is not empty and for every $A, B \in \mathcal{A}$, $|A \cap B| \geqslant l + 1$. Choose any element C from $\mathcal{A}$ and notice that every set in the family must intersect at least with $l + 1$ elements from C, so the number of sets in $\mathcal{A}$ is bounded by

$$|\mathcal{A}| \leqslant \binom{k}{l+1} \binom{n-l-1}{k-l-1} = O(n^{k-l-1}).$$

So we may assume that there are A, B, L as in the hint. Now, either $\mathcal{A} \subset \partial L$ and we are done, or there exists $C \in \mathcal{A}$ such that $L \not\subset C$. How many such C can exists? Because $A \cap B = L$, $L \not\subset C$, $|A \cap C| \geqslant l$ and $|B \cap C| \geqslant l$ we find that

$$|C \cap (A \cup B)| = |C \cap (A \cap B)| + |C \cap (A \setminus B)| + |C \cap (B \setminus A)| \geqslant l + 1$$

where we used the fact that $l \leqslant |C \cap A| = |C \cap L| + |C \cap (A \setminus B)|$ and so $|C \cap (A \setminus B)| \geqslant 1$. So the number of such C is at most $\binom{|A \cup B|}{l+1}\binom{n-|A \cup B|}{k-l-1} = O(n^{k-l-1})$. And now, fix any such C, how many elements of $\mathcal{A}$ may contain L? Because they all intersect in C an element outside of L, we have at most

$$|C \setminus L|\binom{n-l-1}{k-l-1} = O(n^{k-l-1}).$$

such elements, which completes the proof. □

I really liked Rob's motivation for continuing to study these extremal problems.

(a) An explicit construction that gives $R(k) \geqslant k^C$ for any fixed C!

(b) The chromatic number of $\mathbb{R}^n$. We define $\chi(\mathbb{R}^n)$ as the minimal number of colors k such that there exists a function $f : \mathbb{R}^n \to [k]$ satisfying that for every $x, y \in \mathbb{R}^n$ with $|x - y| = 1$, we have that $f(x) \neq f(y)$. For example, we know by taking a simplex on $\mathbb{R}^n$ that $\chi(\mathbb{R}^n) \geqslant n + 1$. It is possible to show (by considering a tesselation by cubes with diameter less than 1) that the number is well defined for any dimension. We will use extremal stuff to prove some order of growth in n of this parameter.

(c) (Borsuk's Conjecture) Show that for every set $X \subset \mathbb{R}^n$ it is possible to decompose it in $n+1$ sets $X_1, \dots X_{n+1}$ s.t. $\operatorname{diam}(X_i) < X$.

Now lets go back to some theorems. The next few ones are nice applications of linear algebra.

Definition 1.6. Given $L \subset [n]$, we say a family $\mathcal{A} \subset \mathcal{P}(n)$ is L-intersecting if for every $A \neq B \in \mathcal{A}$ we have that $|A \cap B| \in L$.

Theorem 1.7. (Modular Frankl-Wilson) Let $\mathcal{A} \subset \mathcal{P}(n)$ and let $L \subseteq \mathbb{Z}_p$ with $|L| = s$. If $\mathcal{A}$ is L-intersecting and for every $A \in \mathcal{A}$, $|A| \pmod p \notin L$, then

$$|\mathcal{A}| \leqslant \sum_{i=0}^{s} \binom{n}{i}$$

Proof. Consider the L.I set

$$f_A(X) = \prod_{l \in L} (X \cdot \mathbf{1}_A - l)$$

in $\mathbb{F}_p[x_1, x_2 \dots x_n]$ and calculate the dimension. □

Theorem 1.8. (Frankl-Wilson) Let $\mathcal{A} \subset \binom{n}{k}$ be L intersecting, then $|\mathcal{A}| \leqslant \binom{n}{\leqslant |L|}$.

Proof. Consider $f_A = \prod_{l \in L}(X \cdot \mathbf{1}_A - l) \in \mathbb{R}[x_1, \dots, x_n]$, this always has degree less than $|L|$ and all monomials (after correcting the exponents) have degree at most $|L|$. So, by the dimension bound (of vectors being L.I), we know that $|\mathcal{A}| \leqslant \binom{n}{\leqslant |L|}$. ☐

Now let's sketch some applications.

Theorem 1.9. There is a explicit coloring that shows $R(m) \geqslant m^C$ for any constant C. (In our setup) There exists a sequence of graphs on N vertices with no clique or independent set of size $N^{o(1)}$.

Proof. The idea is to consider a graph, such that cliques and independent sets correspond to L intersecting families. So let's fix our $V(K_N) = \binom{[n]}{k}$. And consider a coloring of $E[K_N]$ by making

$$c(A, B) = \begin{cases} \text{R} & \text{if} \quad |A \cap B| \in L \\ \text{G} & \text{if} \quad |A \cap B| \notin L \end{cases}$$

where $L \subset [k-1]$. Now, what's the maximum size of a red clique? We know by [1.8], that it is at most by $\binom{n}{\leqslant |L|}$. So if L is relatively small, we can expect to win something. What about green cliques? Then we turn to the modular theorem [1.7]. If we suppose $k = p^2 - 1$ and let $L = \mathbb{Z}_p \setminus \{-1\}$. We would have that each green clique satisfies the hypothesis of 1.7, and so has size at most $\binom{n}{\leqslant p-1}$.

So thats the idea, we take $L = \{p-1, 2p-1, \dots p(p-1)-1\}$ so our coloring has no red clique of size $\binom{n}{\leqslant p-1}$, and no green clique of size $\binom{n}{\leqslant p-1}$.

Suffices to show that if $N = \binom{n}{k}$, then $\binom{n}{\leqslant \sqrt{k}} = N^{o(1)}$. ☐

Theorem 1.10. $\chi(\mathbb{R}^n) \geqslant \exp(cn)$

Proof. The proof follows similarly, we somehow want to find a finite embeddable graph in $\mathbb{R}^n$ with small independence number. Then $\chi(\mathbb{R}^n) \geqslant \chi(G) \geqslant |V(G)|/\alpha(G)$. So, let's look into our setting again of $V(G) = \binom{[n]}{k}$, and to embedd stuff in $\mathbb{R}^n$ we will consider the indicator functions k-sets in $\mathbb{R}^n$ (which are vectors of n dimensions).

Now we want to forbid a distance between these vectors (which will make our sets independent). So let's forbid a specific distance (which we can always scale to be the right thing) and say

$$E(G) = \{(A, B) : |A \cap B| \neq t\}$$

For some $t \in \mathbb{Z}$. Here as we don't really care about the clique number (only the independence), so we need to bound the size of a independent set. So in order to get the strongest bound, we use theorem [1.7] and suppose $k = 2p - 1$ for some p and $L = t \pmod p$. We get $\alpha(G) \leqslant \binom{n}{1}$. So by taking k very big (say $n/2$) we get the result.

☐

Theorem 1.11. (Disproof of Borsuk's Conjecture) $\mathrm{Bor}(\mathbb{R}^n) \geqslant (1+\varepsilon)^{\sqrt{n}}$.

Proof. The proof can be found in Rob's book from 2010 with Imbuzeiro, but follows the same paradigm. ☐

2 Ramsey Theory

Just for sake of completeness, we define

$$R(k,l) = \min\{n : K_k \not\subset G \subset K_n \Rightarrow \alpha(G) \geqslant l\}.$$

In coloring terms, it's the smallest n such that any red and blue coloring of $E(K_n)$ contains a red K_k or a blue K_l.

2.1 Bounds on R(3,k)

We have two very simple bounds that are easy to prove,

$$\left(\frac{k}{\log(k)}\right)^{3/2} \leqslant R(3,k) \leqslant \binom{k+1}{2}.$$

The upperbound follows easily from the Erdös-Szekeres proof.

Theorem 2.1. $R(3,k) \leqslant \binom{k+1}{2}$

Proof. Let G on n-vtx be a triangle-free with no independet set of size k, we know every neighborhood is an independent set so $\Delta(G) < k$. Now we greedily get one vtx at a time removing their neighbors. Pick v_1 and $N(v_1)$ with $|N(v_1)| \leqslant k-1$, then pick v_2, notice that the neighborhood of v_2 (after removing v_1 and its neighbors) has size at most $k-2$, if not $\{v_1\} \cup N(v_2)$ would be a independent set of size k. We proceed like this for $v_1, v_2, \dots v_t$, where each step we discard at most $k-(i-1)$ vtxs. Because we can't do this process k times, for some $i < k$ we must fail (meaning there are no vertex left), so $n = |V(G)| \leqslant \sum_{j=0}^{k-2} k - j \leqslant \binom{k+1}{2} - 1$. □

The lowerbound is a beautiful application of the probabilistic method. The main idea is to choose $G(n,p)$ as to have few triangles (linearly small in n) and relatively low independence number.

Theorem 2.2.

$$R(3,k) \geqslant \left(\frac{k}{\log(k)}\right)^{3/2}$$

We need a **very important** lemma first.

Lemma 2.3. With high probability, the independence number of $G(n,p) \leqslant 2\log(n)/p$. That is

$$\mathbb{P}(\alpha(G(n,p)) \geqslant 2\log(n)/p + 1) = 1 - o_{(n,p(n))}(1)$$

where p can be taken polynomially small in n.

Proof.

$$\begin{aligned}\mathbb{P}(\alpha(G(n,p)) \geqslant k) &\leqslant \mathbb{E}[\#\text{independent k sets} \in G(n,p)] \\ &= \binom{n}{k}(1-p)^{\binom{k}{2}} \leqslant \left(\frac{en}{k}\right)^k \cdot e^{-pk(k-1)/2} \\ &= \left(\frac{en}{e^{p(k-1)/2}k}\right)^k\end{aligned}$$

Taking $k \geqslant 2\log(n)/p + 1$ we get that

$$\mathbb{P}(\alpha(G(n,p)) \geqslant k) \leqslant \left(\frac{ep}{2\log(n)}\right)^{2\log(n)/p} \to 0$$

very fast even! (faster than any polynomial). □

The strategy will be to remove triangles from $G(n,p)$ by removing one vertex from each. This cannot increase the independence number, so we hope the resulting graph still has many vertices to yield a Ramsey construction.

So let's see what this alteration may give us.

Proof. (Of the Theorem) We choose p small such that the number of triangles is no larger than a small linear in n (say $n/2$).

$$\mathbb{E}[\#K_3 \subset G(n,p)] = \binom{n}{3}p^3 = (1+o(1))\frac{(pn)^3}{6}$$

We want

$$\mathbb{P}(\#K_3 \in G(n,p) \geqslant n/2) < (1+o(1))\frac{p^3n^2}{3} < 1/2$$

so taking $p = \varepsilon n^{-2/3}$ for some small ε we get the inequality. But what does this p gives in independence number? We have w.h.p (say bigger than one half)

$$\alpha(G(n,p)) \leqslant 2\varepsilon^{-1}\log(n)n^{2/3}.$$

So $\mathbb{P}(K_3(G(n,p)) < n/2) > 1/2$ and $\mathbb{P}(\alpha(G(n,p) < C\log(n)n^{2/3})) > 1/2$, meaning there exists G satisfying both statements. Remove one vtx from each triangle in G, we are left with a triangle-free graph G' with at least $n/2$ vtxs and $\alpha(G) < C\log(n)n^{2/3}$. Setting $k = C\log(n)n^{2/3}$, we get that

$$V(G') = n/2 \geqslant c\left(\frac{k}{\log(k)}\right)^{3/2}$$

□

2.1.1 Krivilevich lowerbound

Now for a better (and astounishingly beautiful) bound.

Theorem 2.4.

$$R(3,k) \geqslant \left(\frac{k}{\log(k)}\right)^2.$$

The idea of this bound is removing edges. Maybe we can kill the triangles by removing few edges and not increase the independence number too much. For this we will need a beautiful lemma.

Lemma 2.5. (Erdös-Tetali) Let $\{A_1, A_2, \dots A_m\}$ be a family of events in some probability space (Ω, P). Let $\mathcal{E}_t$ be the event that at least t different independent A_i happen. We have that

$$\mathbb{P}(\mathcal{E}_t) \leqslant \frac{1}{t!}\left(\sum_{i=1}^{m} A_i\right)^t.$$

Proof. We get this one from just opening the terms

$$\mathbb{P}(\mathcal{E}_t) = \mathbb{P}\Bigg(\bigcup_{\substack{i_1<\dots<i_t\\ A_{i_j}\not\sim A_{i_k}}} \bigcap_{j=1}^{t} A_{i_j}\Bigg) \leqslant \sum_{\substack{i_1<\dots<i_t\\ A_{i_j}\not\sim A_{i_k}}} \prod_{j=1}^{t} \mathbb{P}(A_{i_j}) \leqslant \sum_{i_1<\dots<i_t} \prod_{j=1}^{t} \mathbb{P}(A_{i_j}) \leqslant \frac{1}{t!}\left(\sum_{i=1}^{m} A_i\right)^t.$$

□

The second ingredient is noting that if you're a bit bigger (say a constant times) than the independent size, you already have many edges (at least half of what is expected).

Lemma 2.6. Let $k \geqslant C\log(n)/p$, then

$$\mathbb{P}\left(\exists \text{ k-set } S, e(G(n,p)[S]) < p\binom{k}{2}/2\right) = o(1)$$

for some $C > 0$.

Proof. Let $\mathcal{E}$ be the event that there exists a k-set S with $e(G(n,p)[S]) < p\binom{k}{2}/2$. We will do a union bound and use Hoeffding (or Chernoff) to bound its probability.

Let S be any k-set. Note that (letting $B(n,p)$ be the binomial with parameters n and p) we have

$$e(G(n,p)[S]) \equiv B\left(\binom{k}{2}, p\right).$$

So by Hoeffding and its bounds on the binomial [A.5] (need to add proofs for these inequalities),

$$\mathbb{P}\left(e(G(n,p)) < \frac{p}{2}\binom{k}{2}\right) = \mathbb{P}\left(\left|B\left(\binom{k}{2}, p\right) - p\binom{k}{2}\right| \geqslant \frac{p}{2}\binom{k}{2}\right) \leqslant e^{-cp\binom{k}{2}}$$

for some $c > 0$.

Now, doing the union bound over k-sets we find that (taking $C > 3c^{-1}$)

$$\mathbb{P}(\mathcal{E}) \leqslant \binom{n}{k} e^{-cp\binom{k}{2}} \leqslant \left(\frac{en}{e^{cp(k-1)/2}k}\right)^k \leqslant \left(\frac{e}{k}\right)^k \to 0.$$

□

The idea to prove theorem 2.4 is to remove a maximal collection of edge-disjoint triangles from $G(n,p)$, (notice that this makes it triangle free). We use lemma 2.5 to show that this procedure won't remove too many edges from any set S with size $C\log(n)/p$.

Proof. (Of Theorem 2.4) Let S be a set of size $k = C\log(n)/p$, for some p we will determine later. Let $\{T_1, \dots T_m\}$ be the set of all triangles that intersect with at least a pair in S. Note that $m \leqslant n\binom{|S|}{2}$. Let $\mathcal{E}_S$ be the event that a edge disjoint set of these triangles with size bigger or equal to $\frac{p}{6}\binom{k}{2}$ are in $G(n,p)$. By [2.5], we can bound $\mathbb{P}(\mathcal{E}_S)$,

$$\mathbb{P}(\mathcal{E}_S) \leqslant \frac{1}{[\frac{p}{6}\binom{k}{2}]!}\left(\sum_{i=1}^{m}\mathbb{P}(T_i \in G(n,p))\right)^t \leqslant \left(\frac{24e}{pk^2}\right)^{pk^2/24}(p^3nk^2)^{pk^2/12}$$

Replacing k in the equation we get (and putting $C > 24e$)

$$\mathbb{P}(\mathcal{E}_S) \leqslant \left(\frac{24ep}{C\log^2(n)}\right)^{\log(n)^2/(24p)}(pn\log(n)^2)^{\log(n)^2/(12p)} \leqslant (p^3n^2\log(n)^2)^{\log(n)^2/(24p)}$$

So, if we take $p = \varepsilon n^{-1/2}$ (might try to optimize more later), we already find that

$$\mathbb{P}(\mathcal{E}_S) \leqslant \left(\frac{\log(n)}{n^{1/2}}\right)^{n^{1/2}\log^2(n)} \ll \left(\frac{C\log(n)}{n^{1/2}}\right)^{C\log(n)n^{1/2}} \leqslant \binom{n}{k}^{-1}$$

So $\mathcal{E}_S$ does not happen for any S with high probability. As w.h.p S of size k have more than $\frac{p}{2}\binom{k}{2}$ edges, and no maximal disjoint triangle family intersect too much with S. We can remove a maximal edge disjoint triangle family and not make any set of size k independent. This yields a triangle free graph on n vtxs with independence number less than $C\log(n)n^{1/2}$, so

$$R(3,k) \geqslant c\left(\frac{k}{\log k}\right)^2.$$

□

2.1.2 AKSz Lemma

Now let's find an upperbound of the right order.

Theorem 2.7. (Ajtai, Komlós, Szemerédi)

$$R(3,k) \leqslant C\frac{k^2}{\log k}$$

The theorem will be immediately implied by the following lemma.

Lemma 2.8. Let G be K_3-free with $\Delta(G) \leqslant d$, then

$$\alpha(G) \geqslant \frac{cn}{d}\log(d).$$

for some universal constant $c > 0$.

Proof. (Of the theorem assuming Lemma 2.8) Notice that G is K_3 free and its independence number is less than k, then, as every neighborhood is a independent set, we have $\Delta(G) < k$. Applying the lemma we get that

$$k > \alpha(G) \geqslant \frac{cn}{k}\log(k)$$

which implies

$$n \leqslant C\frac{k^2}{\log(k)}.$$

□

The proof of the lemma is exceptionally fun. The main idea is to take a uniform random independent set in G, and show that in expectation, its size is bigger than $\frac{cn}{d}\log(d)$ - which is very unnintuitive.

Proof. (Of Lemma 2.8) Let $I \in \mathcal{I}(G)$ be a uniformly chosen random independent set of G. For each $v \in V(G)$, define the random variable

$$X_v = d \cdot \mathbf{1}[v \in I] + \sum_{u \in N(v)} \mathbf{1}[u \in I].$$

Note that

$$\mathbb{E}\sum_{v \in V(G)} X_v = d \cdot \mathbb{E}|I| + \sum_{v \in V(G)}\sum_{u \in N(v)} \mathbb{P}(u \in I) = d \cdot \mathbb{E}|I| + \sum_{u \in V(G)} d(u)\mathbb{P}(u \in I) \leqslant 2d \cdot \mathbb{E}|I|.$$

So we have that $\mathbb{E}|I| \geqslant \frac{1}{2d}\mathbb{E}[\sum_{v \in V(G)} X_v]$. To prove the lemma, it would suffice to show then that $\mathbb{E}X_v \geqslant \log(d)$ for all v (as summing it would yield the result).

So let's lowerbound $\mathbb{E}X_v$. The idea is to condition it on something and find a bound that works **FOR EVERY** condition we take (which seems insane).

Fix v, let's divide the world into $U = \{v\} \cup N(v)$ and $W = V(G) \setminus U$ (both depend on v). Notice that the value of X_v only changes for elements in U that appear in I. So fix an independent set S of W, and let's calculate $\mathbb{E}[X_v | I \cap W = S]$.

From the fact that I has to be an independent set, we know it cannot have any neighbors of elements in S. So let $Y = N(v) \cap N(S)$ and let $X = N(v) \setminus Y$ (note $v \notin N(S)$). As I is uniformly chosen, we now know that $I \cap U$ (conditioned on $I \cap W = S$), is a uniformly chosen independent set of $\{v\} \cup X$. Let $|X| = t$, remembering that X is an independent set, we find

$$\begin{aligned}\mathbb{E}[X_v | I \cap W = S] &= \mathbb{E}[X_v | v \in I, I \cap W = S] \cdot \mathbb{P}(v \in I | I \cap W = S)\\ &\quad + \mathbb{E}[X_v | v \notin I, I \cap W = S] \cdot \mathbb{P}(v \notin I | I \cap W = S)\\ &= \frac{1}{2^t+1}d + \frac{2^t}{2^t+1}\frac{t}{2}\end{aligned}$$

where we used the fact that $I \cap U$ is an uniform independent set of the star $G[\{v\} \cup X]$. Now (here comes the crazy part), notice that independent of what $t = |X|$ is, we have

$$\frac{1}{2^t+1}d + \frac{2^t}{2^t+1}\frac{t}{2} \geqslant \frac{\log(d)}{8}.$$

just take $t < \log(d)/2$ and $t \geqslant \log(d)/2$. But this means

$$\mathbb{E}X_v = \mathbb{E}[\mathbb{E}[X_v | I \cap W(v)]] \geqslant \frac{\log(d)}{8}$$

yielding the result. □

2.2 Bounds on R(4,k)

By doing vertex deletion and Erdös-Szekeres, we may easily find (we need $p^6n^4 < n$, so $p = \varepsilon n^{-1/2}$)

$$c\left(\frac{k}{\log(k)}\right)^2 \leqslant R(4,k) \leqslant k^3.$$

Doing the Erdös-Tetali method [2.5] (we need $p^6n^4 < pn^2$, so $p = \varepsilon n^{-2/5}$) we get also

$$c\left(\frac{k}{\log k}\right)^{5/2} \leqslant R(4,k).$$

But we can do better on the upperbound by doing a similar strategy as in the $R(3,k)$ case. The next lemma Rob stated, but I don't think we will use it.

Lemma 2.9. Let G on n vtx be K_4-free and $\Delta(G) \leqslant d$, then

$$\alpha(G) \geqslant \frac{cn}{d}\frac{\log(d)}{\log\log(d)}.$$

Proof. The proof of this is similar to lemma 2.8, we must however pick a uniformly independent set from the now K_3-free X. (I will try to do the computations more carefully later) □

To get an even better bound, we will need some kind of stability setting in these super sparse graphs.

Lemma 2.10. (AKSz) Let G be a graph on n-vtx with $\Delta(G) \leqslant d$ and $k_3(G) \leqslant (nd^2)/\lambda$, then

$$\alpha(G) \geqslant c\frac{n}{d}\log(\lambda)$$

where $c > 0$ is an absolute constant.

With it we can prove the following strong bound,

Theorem 2.11.

$$R(4,k) \leqslant C\frac{k^3}{\log(k)^2}.$$

Proof. (Assuming [2.10]) Let G be n-vtx, K_4-free graph with $\alpha(G) < k$, we know by Erdös-Szekeres that $\Delta(G) < R(3,k)$. How many triangles can G have? Pick a vertex v and let $S = N(v)$. We know that for any vertex $u \in S$, $N_{G[S]}(u)$ is independent (because G is K_4-free) and so $d_{G[S]}(u) < k$. This implies that

$$k_3(G) = \frac{1}{3}\sum_{v \in V(G)} e(G[N(v)]) \leqslant \frac{nk^2}{6}$$

by our bound on $\Delta(G[S])$.

So applying [2.10] with $d = R(3,k)$ and $\lambda = (k/\log(k))^2$ yields

$$k > \alpha(G) > c\frac{n\log(k)^2}{k^2}$$

which yields $n < Ck^3/\log(k)^2$. □

So now it suffices to prove 2.10.

Proof. (Of the AKSz lemma) The idea will be to do some vertex deletion and apply 2.8, but we don't really know how so we do it randomly.

To easy up the calculation (it only changes the constant in front), suppose $\Delta(G) \leqslant d$ and $k_3(G) \leqslant nd^2/\lambda^3$. Take a q-random subset Q of $V(G)$, we can estimate some statistics, we find:

(i) $\mathbb{E}|Q| = qn$

(ii) $\mathbb{E}e(G[Q]) = q^2 e(G) \leqslant q^2 nd/2$

(iii) $\mathbb{E}k_3(G[Q]) = q^3 k_3(G) \leqslant (q/\lambda)^3 nd^2$

By taking $q = \lambda/d$ and using Markov to remove triangles and high degree vtxs, we find a set of size at least $qn/6$ with no triangles and $\Delta(G[Q']) < 3q\Delta(G)$. Applying 2.8 to this set yields the result.

(Formally) We have very good concentration on $|Q|$ (as long as $qn \to \infty$), so applying Chernoof and Markov (twice) we see that

$$\begin{aligned}
\mathbb{P}\Bigg(\Big(|Q| > \frac{qn}{2}\Big) \wedge \Big(e(G[Q]) < 3q^2 e(G)\Big) \wedge \Big(k_3(G[Q]) < \frac{3q^3nd^2}{\lambda^3}\Big)\Bigg) \\
&= 1 - \mathbb{P}\Bigg(\Big(|Q| < \frac{qn}{2}\Big) \vee \Big(e(G[Q]) > 3q^2 e(G)\Big) \vee \Big(k_3(G[Q]) > \frac{3q^3nd^2}{\lambda^3}\Big)\Bigg) \\
&\geqslant 1 - o(1) - 1/3 - 1/3 > 0
\end{aligned}$$

So we find a $G' = G[Q]$ for some good Q satisfying all these properties. G' satisfies:

(i) $v(G') \geqslant \frac{qn}{2}$

(ii) $e(G') \leqslant 3q^2 e(G) \leqslant 3q^2 nd/2$

(iii) $k_3(G') \leqslant \frac{3q^3nd^2}{\lambda^3}$

By removing at most half of the vertices of G' (using Markov) and killing a vertex from each triangle, we find triangle free G'' with $v(G') \geqslant \frac{qn}{4} - k_3(G') \geqslant \frac{qn}{8}$ and $\Delta(G'') \leqslant 6q^2nd/2$. ☐

This seems like a very interesting strategy for off-diagonal Ramsey. We did the case $l = 3$, $l = 4$, so a very natural conjecture would be:

Theorem 2.12.

$$R(l, k) \leqslant C\frac{k^{l-1}}{(\log k)^{l-2}}$$

Proof. We prove it by induction. The case $l = 3, 4$ were already done. Let's follow the proof of [2.11]. Suppose G is an n-vtx, K_l-free graph with $\alpha(G) < k$. We must have $\Delta(G) < R(l-1, k) \leqslant d$. How many triangles can it have? We know that if $u \in N(v)$, then $|N(u) \cap N(v)| < R(l-2, k)$. So, doing the same bound as before, we see that $k_3(G) \leqslant ndR(l-2, k)/3$. So, to use 2.10 we must have

$$\frac{ndR(l-2,k)}{3} \leqslant \frac{nd^2}{\lambda}$$

so

$$\lambda \leqslant \frac{3d}{R(l-2,k)}$$

and (by induction), it suffices that λ satisfies

$$\lambda \leqslant \frac{cd(\log k)^{l-2}}{k^{l-1}}$$

applying 2.10 with $d = \frac{k^{l-1}}{(\log k)^{l-2}} \geqslant R(l-1,k) > \Delta(G)$, we have by hypohesis,

$$\alpha(G) \geqslant \frac{n}{d}\log(\lambda) \geqslant C_l \frac{n\log(k)^{l-2}}{k^{l-2}}$$

and therefore,

$$n \leqslant C_l \frac{k^{l-1}}{(\log k)^{l-2}}$$

□

This proof can also be found in Rob's ICM manuscript. [https://arxiv.org/pdf/2601.05221].

2.3 Bounds on R(3,3,k)

So what are some easy bounds we can get on this? An easy lowerbound is $R(3,k)$, by doing Erdös-Szekeres, we can get the multinomial bound as well

$$R(3,k) \leqslant R(3,3,k) \leqslant 2kR(3,k) \leqslant C\frac{k^3}{\log k}.$$

We can do a bit better on the upperbound by counting triangles and applying [2.10]!

Theorem 2.13.

$$R(3,3,k) \leqslant \frac{k^3}{(\log k)^2}$$

Proof. Suppose there is a coloring χ on K_n that shows that $R(3,3,k) \geqslant n$. Let $G = E_R(\chi(K_n)) \cup E_B(\chi(K_n))$ that is, G is the graph of red and blue edges. Note that $\alpha(G) < k$, as the coloring had no green K_k. To apply [2.10], we must calculate $k_3(G)$ and we know it does not have any full red, or any full blue triangles. So let's bound the number of $\{R,R,B\}$ triangles in G (the number of $\{B,B,R\}$ by simmetry will be the same). One way to count $\{R,R,B\}$ triangles is by picking the cherry vertex.

$$k_{\{R,R,B\}}(G) = \sum_{v\in G} e_B(G[N_R(v)])$$

By Erdös-Szekeres, we know that $N_R(v)$ is red-edge-free and $|N_R(v)| \leqslant R(3,k)$ for any vertex v. We also have that for any $u \in N_R(v)$, $N_B(u) \cap N_R(v)$ is a independent set, by the previous observation and the fact that we are blue triangle free. So $d_B[G[N_R(v)]](u) < k$ (here this monstrosity means that the blue degree of u in $N_R(v)$ is less than k). This implies that $e_B(G[N_R(v)]) < R(3,k)k/2$ for any v and therefore that

$$k_{\{R,R,B\}}(G) < \frac{nR(3,k)k}{2}$$

so the total number of triangles is also of order $nk^3/\log k$. Applying [2.10] with $d = \frac{Ck^2}{\log k} \geqslant R(3,k)$ and this $k_3(G) < ndk$ we get that we can use $\lambda = Cd/k \geqslant \frac{Ck}{\log k}$. So we get

$$k > \alpha(G) \geqslant \frac{n}{d}\log k \geqslant \frac{C'n(\log k)^2}{k^2}$$

which implies $n \leqslant Ck^3/(\log k)^2$. □

Next class we will see a very cool lowerbound by Alon, who observed the following lemma:

Lemma 2.14. If there exists K_3-free G on n vtx with number of independent k sets of G less than $\binom{n}{k}^{1/2}$ then $R(3,3,k) > n$.

Proof. Embed G randomly in K_n with color red, do it again in color blue. By the union bound, no independent k set survive (so we win by painting non edges with green). □

We will turn for a pseudo-random construction to prove the existence of a graph that satisfies [2.14]. For this, we need a good definiton of what pseudo-random means.

Definition 2.15. We say that a n-vtx graph G is (p,β)-jumbled if $e(G) = p\binom{n}{2}$ and for every $S,T \subset V(G)$, we have that

$$|e(S,T) - p|S||T|| \leqslant \beta\sqrt{|S||T|}.$$

where here $e(S,T)$ is the number of edges that start in S and end in T. Edges in $S \cap T$ are counted twice.

The most pseudo-random graph (the best β you can get) is with $G(n,p)$, I don't know anyway to prove this now, but let's calculate how jumbled $G(n,p)$ is.

Theorem 2.16. $G(n,p)$ is (p',β)-jumbled with $p' = e(G(n,p))/\binom{n}{2} = (1+o(1))p$ and $\beta = O(\sqrt{pn})$ w.h.p.

Proof. We use Chernoff inequality [A.5], to note that for absolute $C,c > 0$ and taking A suffciently big,

$$\mathbb{P}(|e(S,T) - p|S||T|| \geqslant A\sqrt{pn|S||T|}) \leqslant C\exp(-cA^2 n) \ll 4^{-n},$$

so $G(n,p)$ is $\beta = A\sqrt{np}$-jumbled w.h.p by the union bound. □

This yields the following definition

Definition 2.17. A graph is said to be **optimally pseudorandom** if it is (p,β)-jumbled with $\beta = O(\sqrt{pn})$.

The following question is at the core of this method. How many independent k-sets can a (p,β)-jumbled graph have? We now have a deterministic fact about our graph and we can try to count. I won't do it here, but the idea is to count sequences $(v_1,\ldots,v_k)$ of vertices that can make a independent set.

Using the (p,β) jumbled condition, we can count how many "bad" vertices there are when trying to grow the sequence. We can make only a few "good" steps (extension steps where our non-neighborhood decreases with correct size), because they reduce the non-neighborhood by a $(1-p)/2$ ratio which is just too fast. And each "bad" step (steps which don't reduce the non-neighborhood much) we are obliged to choose from a small set. So counting everything yields a result.

The following lemma will be stated without proof, it is an algebraic construction.

Lemma 2.18. (Quadrangle) There exists a n-vertex, $n^{1/3}$-regular, $n^{1/3}$-uniform hypergraph H that satisfies:

(i) $\forall e_1, e_2 \in E(H), \quad |e_1 \cap e_2| \neq 2.$

(ii) $\forall e_1, e_2, e_3 \in E(H)$ it does not hold that $|e_1 \cap e_2| = |e_2 \cap e_3| = |e_1 \cap e_3| = 1.$

Putting a randomly embedded $K_{t,t}$ with $t = n^{1/3}/2$ on each edge we get a K_3-free, optimally pseudorandom graph where p is the maximum it can be, $p = n^{-1/3}$. (As Rob pointed out an optimally pseudorandom, triangle-free graph must have density at most $p = n^{-1/3}$).

2.4 Ramsey of bounded degree graphs

The motivation of this section will be the following theorem.

Theorem 2.19. Let H be a bipartite graph with k vertices and $\Delta(H) \leqslant d$. Then $r(H) \leqslant 2^{d+6}k$.

2.4.1 Old Friends

To uncover it, we will need to remember some old forgotten friends. The LLL and the DRC.

Theorem 2.20 (Lovász Local Lemma)**.** Let $\mathcal{E} = \{B_1, B_2, \ldots, B_m\}$ be bad events in some probability space and suppose that for each event B_i, there exists a set $N(B_i) \subset \mathcal{E}$ such that B_i is mutually independent of all events in $\mathcal{E} \setminus N(B_i)$. We say that $j \sim i$ if $B_j \in N(B_i)$. If there exists $x_1, \ldots x_m \in (0,1)$ satisfying

$$\mathbb{P}(B_i) \leqslant x_i \prod_{j\sim i}(1-x_j)$$

then

$$\mathbb{P}\left(\bigcap_{i=1}^{m} B_i^c\right) > \prod_{i=1}^{m}(1-x_i).$$

Remark 2.21. Usually people talk about the dependence graph, in this framework the neighbors of an event B_i are precisely $N(B_i)$. Note that the choice of $N(B_i)$ (or of the dependece graph) is not necessairly unique.

Proof of LLL 2.20. The proof is just a very smart induction (similar to Harris or Janson - which I should also add in the appendix). We claim that given any $S \subset [m]$,

$$\mathbb{P}\left(B_i \mid \bigcap_{j \in S} B_j^c\right) \leqslant x_i.$$

Note that this implies the theorem as we can always write

$$\mathbb{P}\left(\bigcap_{i=1}^{m} B_i^c\right) = \prod_{i=1}^{m} \mathbb{P}\left(B_i^c \mid \bigcap_{j<i} B_j^c\right)$$

and the result follows from the claim.

Let's prove the claim by induction on $|S|$. If $S = \varnothing$, it trivially follows as $\mathbb{P}(B_i) \leqslant x_i \prod_{j \sim i}(1 - x_j) < x_i$. Now let's separate the event

$$\bigcap_{j \in S} B_j^c = \bigcap_{\substack{j \in S \\ j \sim i}} B_j^c \cap \bigcap_{\substack{j \in S \\ j \not\sim i}} B_j^c = E \cap F$$

we think of E as the influence of the "neighbors" in the dependence graph and F as the influcence of the non-neighbors. We can now write

$$\mathbb{P}(B_i \mid E \cap F) = \frac{\mathbb{P}(B_i \cap E \cap F)}{\mathbb{P}(E \cap F)} = \frac{\mathbb{P}(E \mid B_i \cap F)\mathbb{P}(B_i \mid F)\mathbb{P}(F)}{\mathbb{P}(E \mid F)\mathbb{P}(F)} \leqslant \frac{x_i \prod_{j \sim i}(1 - x_j)}{\mathbb{P}(E \mid F)} \tag{1}$$

If we bound $\mathbb{P}(E \mid F) \leqslant 1$ and use the independence to say $\mathbb{P}(B_i \mid F)$ is less than what given by the hypothesis. Now what is $\mathbb{P}(E \mid F)$? Opening it up we get (we abuse the notation and say that $j \in E$ if E is taking the intersection with B_j^c)

$$\mathbb{P}(E \mid F) = \prod_{j \in E} \mathbb{P}\left(B_j^c \mid \bigcap_{\substack{k<j \\ k \in E}} B_k^c \cap F\right) \geqslant \prod_{j \in E}(1 - x_j)$$

by the induction hypothesis. Substituting in the inequality [1] we prove the claim. □

Remark 2.22. We cannot take the graph to be degenerate in the symmetric case as we need the I.H. to compute $\mathbb{P}(E \mid F)$.

Remark 2.23. In the hypothesis, we didn't use full mutual independence, only that $\mathbb{P}(B_i \mid F) \leqslant \mathbb{P}(B_i)$. Using only this requirement, we call it *Lopsided LLL.*

Corollary 2.24 (Symmetric LLL)**.** Suppose that in the hypothesis of [2.20], we had $\max_i |N(B_i)| \leqslant d$. That is the dependence graph D is such that $\Delta(D) \leqslant d$. Suppose moreover that for all i,

$$\mathbb{P}(B_i) \leqslant \frac{1}{e(d+1)}$$

then $\mathbb{P}(\bigcap_{i \leqslant m} B_i^c) > 0$.

Proof. Take $x_i = 1/(d+1)$, and note that

$$\mathbb{P}(B_i) \leqslant \frac{1}{e(d+1)} \leqslant \frac{1}{(d+1)}\left(1 - \frac{1}{d+1}\right)^d \leqslant x_i \prod_{j\sim i}(1-x_j).$$

□

The LLL is a pearl that should always be within our hearts. But there is a even simpler theorem which we must never forsake.

Theorem 2.25 (Dependent Random Choice)**.** Let $k, s \in \mathbb{N}$ and G be any graph with n vtxs. For any $t \in \mathbb{N}$, there exists a set $U \subset V(G)$ with size greater or equal to

$$\frac{(2e(G))^t}{n^{2t-1}} - \binom{n}{k}\left(\frac{s}{n}\right)^t$$

such that every set of k vertices of U has s neighbors in common.

Proof. Choose independently and uniformly t vertices from $V(G)$ with possible repetition. Call this random set T and let $X = |N(T)|$. Let Y be the number of k-sets in $N(T)$ with less than s common neighbors. Let's compute $\mathbb{E}X$ and $\mathbb{E}Y$. We may write

$$X = \sum_{v\in V(G)} \mathbf{1}[v \in N(T)]$$

So $\mathbb{E}X = \sum_{v\in V(G)} \mathbb{P}(v \in N(T))$. Now if $v \in N(T)$, then every vertex in T belongs to $N(v)$, as we chose u.i.i.d we find

$$\sum_{v\in V(G)} \mathbb{P}(v \in N(T)) = \sum_{v\in V(G)} \mathbb{P}(T \subset N(v)) = \sum_{v\in V(G)} \left(\frac{d(v)}{n}\right)^t \geqslant n\left(\frac{2e(G)}{n^2}\right)^t$$

where the last inequality follows by Jensen.

Now what about Y? To ease notation, let (only this once) $(V(G))_{[k]} = \binom{V(G)}{k}$, then

$$Y = \sum_{\substack{S\in(V(G))_{[k]}\\ |N(S)|<s}} \mathbf{1}[S \subset N(T)] = \sum_{\substack{S\in(V(G))_{[k]}\\ |N(S)|<s}} \mathbf{1}[T \subset N(S)]$$

taking expectation and upperbounding by all k-sets, we get

$$E[Y] \leqslant \binom{n}{k}\left(\frac{s}{n}\right)^t.$$

We have that

$$\mathbb{E}[X - Y] \geqslant \frac{(2e(G))^t}{n^{2t-1}} - \binom{n}{k}\left(\frac{s}{n}\right)^t = a \tag{2}$$

so there exists an instance of T, such that by removing a vertex from each bad k-set in $N(T)$, we are still left with a vertices. □

2.4.2 Applications

Let's use the LLL [2.20] to prove lowerbounds on Ramsey.

Erdös showed that, in his majestic use of the probabilistic method, the following theorem.

Theorem 2.26. Consider a random coloring of the edges of K_N in red and blue, each chosen independently and uniformly. If X is the number of monochromatic cliques of size k and $\mathbb{E}X < 1$, then $R(k) > N$. In particular, $R(k) \gtrsim 2^{k/2}$.

Proof. The first statement is a proof in itself, maybe the most beautiful proof of all. For the second we only need to do the computation. We know

$$\mathbb{E}X = \binom{n}{k} 2^{-\binom{k}{2}+1} \leqslant \left(\frac{en}{k} 2^{-k/2}\right)^k.$$

(We lose a bit when doing the binom approximation and we use it). By choosing n appropriatly, this yields

$$R(k) > \frac{k}{e} 2^{k/2}.$$

□

Spencer got a crazy improvement, unbeaten by the ages (50 years already), using the LLL.

Theorem 2.27. (Spencer 77)

$$R(k) > \frac{2k}{e} 2^{k/2}.$$

He also managed to prove Erdös lowerbound on $R(3, k)$ in a much simpler way - way before Krivilevich's proof [see Theorem 2.4].

2.4.3 Proof of Conlon-Fox-Sudakov

We will now try to prove Theorem [2.19] using a twisted combination of our old friends. It's a very nice proof.

The first method that comes to mind when trying to find bipartite graphs is DRC, but it alone won't be strong enough to give us the upperbound we want. So let's do something a bit better.

Lemma 2.28. Let G be a graph on n vtxs and suppose $e(G) \geqslant c\binom{n}{2}$ for some $c \leqslant 1/2$. Suppose moreover that $d, k \in \mathbb{N}$ are such that

$$2d \leqslant k \leqslant c^{d+2} n.$$

Then, there exists a set $A \subset V(G)$ satisfying

(a) $|A| \geqslant k$

(b) At most $c^d \binom{|A|}{d}$ d-subsets of A have less than k neighbors in common.

Proof. The proof of this will be exactly the same as the proof of DRC [2.25] but we will take expectation over a different expression [2]. Let $T = (u_1, \dots u_t)$ be a random uniformly independently chosen set of vertices. As before we set $X = |N(T)|$, and Y the number of bad d-sets. Again we have that

$$\mathbb{E}X \geqslant c^t n \quad \text{and} \quad \mathbb{E}Y \leqslant \binom{n}{d}\left(\frac{k}{n}\right)^t$$

Crucially, we want to show that with positive probability $X \geqslant k$ and $Y \leqslant c^d\binom{|X|}{d}$. So we will take the mean of a magical r.v with positive expectation. Consider Z given by

$$Z = X^t - \frac{(\mathbb{E}X)^t}{2} - \frac{(\mathbb{E}X)^t}{2}\frac{Y}{\mathbb{E}Y}.$$

Note that as $X, Y \geqslant 0$, we know that $\mathbb{E}X^t \geqslant (\mathbb{E}X)^t$, so

$$\mathbb{E}Z = \mathbb{E}X^t - \frac{\mathbb{E}[X]^t}{2} - \frac{\mathbb{E}[X]^t}{2} \geqslant 0.$$

As both $(\mathbb{E}X)^t/2$ and $(\mathbb{E}X)^t Y/(2\mathbb{E}Y)$ are positive, it follows that with positive probability $X^t \geqslant (\mathbb{E}X)^t/2$ and, most importantly,

$$X^t \geqslant \frac{(\mathbb{E}X)^t}{2}\frac{Y}{\mathbb{E}Y}.$$

So we found that with probability > 0,

$$Y \leqslant \binom{n}{d}\left(\frac{k}{n}\right)^t \frac{2X^t}{c^{t^2}n^t} \leqslant \binom{n}{d}\frac{2c^{t(d+2)}X^t}{c^{t^2}n^t}$$

taking $t = d$, we get

$$Y \leqslant \frac{c^{2d}X^d}{d!} \leqslant c^d\binom{X}{d}$$

if d is big enough. The other bound follows easily,

$$X \geqslant \frac{\mathbb{E}X}{2^{1/d}} \geqslant \frac{c^d n}{2^{1/d}} \geqslant k.$$

□

With this new lemma, let's prove a weaker version of [2.19] (maybe I can see later how the stronger one follows.)

Theorem 2.29. Let H be a d-regular bipartite graph on k vtxs. Then

$$r(H) \leqslant 2^{d+5}k.$$

Proof. Let $n \geqslant 2^{d+5}k$ and G a graph on n vtxs with $e(G) \geqslant \frac{1}{2}\binom{n}{2}$ - this is a density proof, actually we find H in the densest color. Suppose $H = A \sqcup B$ and apply lemma [2.28] to G with $c = 1/2$. We find a set A^* satisfying the conclusions of the lemma. We will use the LLL [2.20] to randomly embed A into A^* in such a way that for every $b \in B$, $N(b)$ is sent to a good set in A^*. Because of this, we can trivially find a copy of H in G.

To really use the LLL, we cannot actually embed it ramdomly, this would kill all the mutual independence of the events. What we actually do is a random assignment.

Let $f : A \to A^*$ be a random map. We will define two types of bad events, collisions and bad sets. For each $u, v \in A$, define the events

$$F_{u,v} = \{f(u) = f(v)\}$$

and for each b in B, define

$$E_b = \{f(N(b)) \text{ is a bad } d\text{-set}\}.$$

Amazingly here, if non of the $F_{u,v}$ occur, no neighborhood is mapped to less than d elements, so we don't even need to care that $f(N(b))$ has the right size. Now for the dependence graph, by how we chose our probability space, we have that disjoint events (on the vertex of H) are non-edges:

- $F_{u,v} \sim F_{u',v'}$ if $\{u, v\} \cap \{u', v'\} \neq \varnothing$.
- $F_{u,v} \sim E_b$ if $\{u, v\} \cap N(b) \neq \varnothing$.
- $E_b \sim E_{b'}$ if $N(b) \cap N(b') \neq \varnothing$.

We need to find x_F and x_E satisfying the hypothesis of 2.20. We know that

$$\mathbb{P}(F_{u,v}) = \frac{1}{|A^*|} \leqslant \frac{1}{k}$$

and

$$\mathbb{P}(E_b) \leqslant c^d.$$

By counting the neighborhoods of the events, we need to satisfy

$$\begin{aligned}\mathbb{P}(E_b) &\leqslant c^d \leqslant x_E(1 - x_E)^{d^2}(1 - x_F)^{d|A|} \\ \mathbb{P}(F_{u,v}) &\leqslant \frac{1}{|A^*|} \leqslant x_F(1 - x_E)^{2d}(1 - x_F)^{2|A|}\end{aligned}$$

so we really should check which is biting us by trying both x_E close to $\mathbb{P}(E_b)$ or x_F close to $\mathbb{P}(F_{u,v})$. In this case, the collisions are biting. Let $x_F = 2/|A^*|$, then (we might have to take A^* bigger here this is why we get a 2^{d+6}, we do $|A^*| = 8|A|$)

$$\frac{1}{|A^*|} \leqslant \frac{2}{|A^*|}(1 - x_E)^{2d} e^{-2|A|/|A^*|},$$

So we need

$$(1 - x_E)^{2d} \geqslant \frac{1}{2e^{1/8}}.$$

From the first inequality we also get

$$c^d \leqslant x_E(1 - x_E)^{d^2} e^{-d|A|/|A|^*} \leqslant x_E e^{-x_E d^2} e^{-d/8}$$

so taking $x_E = (ce^{1/8})^d$ we get the result when d tends to infinity.

□

3 The Method of Containers

3.1 Some nice problems

There are many reasons to understand the method of hypergraph containers. Here are some cool problems we may be interested in.

Problem 3.1. (Extremal questions in $G(n,p)$). We define

$$\mathrm{ex}(G,H) := \max\{e(G') : G' \subset G,\ G \text{ is } H-\text{free}\}.$$

Note that $\mathrm{ex}(n,H) = \mathrm{ex}(K_n,H)$. What can we say about $\mathrm{ex}(G(n,p),K_3)$?

Some easy bounds can be found. Given any sampling of $G(n,p)$, we can always find a bipartite graph with half the edges. Similarly another strategy is just to kill off triangles by removing an edge from each. This already gives

$$\mathrm{ex}(G(n,p),K_3) \geqslant \begin{cases} e(G(n,p))/2 \\ e(G(n,p)) - k_3(G(n,p)) \end{cases}$$

by finding the best regime for p in each we get

$$\mathrm{ex}(G(n,p),K_3) \geqslant \begin{cases} pn^2/4 \quad \text{for any}\, p \in [0,1] \\ (1-o(1))pn^2/2 \quad \text{for}\, p \ll \frac{1}{\sqrt{n}} \end{cases}$$

it turns out this is sharp.

Problem 3.2. When does $G(n,p) \to K_3$? Define

$$p^*(n) = \inf_{p\in[0,1]} \{p : \mathbb{P}(G(n,p) \to K_3) \geqslant 1/2\}.$$

can we find this threshold?

We know that if $G(n,p)$ contains a K_6 we are done by Ramsey so $p^* \lesssim n^{-2/5}$. But is this it? Should avoiding K_6 be enough or is there some deeper more complex structure - the answer is obviously yes.

Problem 3.3. Let $A \subset [n]_p$ with $|A| \geqslant \delta np$, does A contain a 3-AP?

Doing deletion we get that it is true if $AP_3([n]_p) = o(np)$. We know $AP_3([n]_p) \approx n^2p^3$, so if $p \ll 1/\sqrt{n}$ we get the result. What happens if $p \gg 1/\sqrt{n}$?

Very related to 3.4 is question:

Problem 3.4. How many K_3-free graphs on n vertices with m edges are there?

If we call this quantity $X(n,m)$, we know (because we can always pick from a bipartite graph) that

$$X(n,m) \geqslant \binom{n^2/4}{m}.$$

If m is sufficiently low, we can use $G(n,p)$ for our advantage! We know that

$$\mathbb{P}(K_3 \not\subset G(n,p)) \approx \exp(-n^3p^3)$$

if $p \ll n^{-1/2}$ by Janson and Harris inequality. Noting that $G(n,m) \approx G(n,p)$ with $p = m/\binom{n}{2}$, we get something that says

$$X(n,m) \approx \exp(-m^3/n^3)\binom{\binom{n}{2}}{m} \quad \text{if } m \ll n^{3/2}.$$

Is the first bound optimal?

3.2 Attempts at handling them

Let's follow a naive approach for tackling 3.1. So let's see first moment. Let

$$Y = \#\big\{G : K_3 \not\subset G \subset G(n,p) \text{ with } e(G) > m = (1/2+\varepsilon)p\binom{n}{2}\big\}$$

the number of such bad subgraphs of $G(n,p)$. We would like to show $Y = 0$ with high probability if $p \gg 1/\sqrt{n}$. If we calculate $\mathbb{E}Y$ we get

$$\mathbb{E}Y = \sum_{\substack{K_3 \not\subset G \\ e(G)=m}} \mathbb{P}(G \subset G(n,p)) \geqslant \binom{n^2/4}{m}p^m \approx \left(\frac{en^2p}{4m}\right)^m \approx e^m$$

where for the lowerbound we only considered the subsets of a single bipartite graph. But $\mathbb{E}Y$ is **HUGE**, so can we better this?

A very interesting point made by Rob is to look where we are losing here. Every once and while, when you look at an equation and it explodes, it is a good idea to see why. Maybe we could handle the bad case differently.

Here we are losing a ton because there are many subsets of the bipartite graph, but this seems a bit unrealistic, a bipartite graph contains at most $1/2$ of the edges of $G(n,p)$ and here we are asking for an ε deviation. To be more precise, let's check another thing.

What can we say about bipartite H in $G(n,p)$ with m edges? Can we find many of those? Note that every bipartite graph H is a subset of a complete bipartite G which has at most $n^2/4$ edges. So we could say

$$\mathbb{P}(\exists \text{ bip.}\, H \subset G(n,p) \text{ with } e(H) = m) = \sum_{\substack{K_{(n/2,n/2)} \sim G \\ G \subset G(n,p)}} \mathbb{P}(\exists \text{ bip.}\, H \subset G \cap G(n,p) \text{ with } e(H) = m)$$

It might look that we made our lives worse, but in reality this is a way simpler thing. For such H to exists, $e(G \cap G(n,p)) \geqslant m$, so here we can already apply a deviation bound. Furthermore, the number of complete bipartites is 2^{n-1} so applying the union bound seems like a great idea. We get (using A.5)

$$\sum_{\substack{K_{(n/2,n/2)}\sim G\\ G\subset G(n,p)}} \mathbb{P}(\exists \text{ bip. } H\subset G\cap G(n,p) \text{ with } e(H)=m) \leqslant 2^n\mathbb{P}(\mathrm{Bin}(n^2/4,p)\geqslant m)$$
$$\leqslant 2^n\exp(-\varepsilon^2 n^2 p/4)\to 0$$

if $p \gg 1/n$ so we win by miles in this bipartite case.

What we actually did here is a container theorem for bipartite graphs. Let $\mathcal{G}$ be the family of complete bipartite graphs on n vtxs, we know that

(a) $|\mathcal{G}|$ is small $\leqslant 2^n$.

(b) Every $G\in\mathcal{G}$ is bipartite.

(c) For every bipartite $H\subset K_n$, there exists $G\in\mathcal{G}$ such that $H\subset G$. (G is the container for H).

This family allowed us to do way more efficient union bound. And this is what will try to do to tackle 3.1.

For our problem, this we need another container theorem which we will prove later:

Theorem 3.1. (Container theorem for K_3-free graphs). There exists a family $\mathcal{G}$ of graphs on n-vtxs, s.t

(a) $\mathcal{G}$ is small, $|\mathcal{G}|\leqslant 2^{n^{3/2}(\log n)^c}$.

(b) Every $G\in\mathcal{G}$ has "few" triangles, $o(n^3)$.

(c) For every K_3-free graph H, there exists a container $G\in\mathcal{G}$ s.t. $H\subset G$.

An answer for 3.1. Note that we can upperbound $\mathbb{P}(\exists K_3\not\subset H\subset G(n,p), e(H)=m)$ using our containers. How many edges can our containers have? Because they have $o(n^3)$ triangles, by supersaturation they must have at most $(1/2+o(1))\binom{n}{2}$ edges. So following the same calculation as before (doing the union bound and applying our deviation result), we get

$$\mathbb{P}(\exists K_3\not\subset H\subset G(n,p), e(H)=m)\leqslant \sum_{G\in\mathcal{G}}\mathbb{P}(e(G\cap G(n,p))\geqslant m)\leqslant 2^{n^{3/2}\log(n)^c}e^{-\varepsilon pn^2}\to 0$$

if $p\gg \log(n)^c/\sqrt{n}$. □

Next time we will try to prove Theorem 3.1, and we will do so in greater generality. We will do it for 3-uniform hypergraphs and apply it for a "triangle encoding" hypergraph.

Definition 3.2. The triangle enconding hypergraph $\mathcal{H}_{K_3}$ is a 3-uniform hypergraph with $V(\mathcal{H}_{k_3}) = E(K_n)$ and $E(\mathcal{H}_{k_3}) = \{\{e,f,g\} : e,f,g\in E(K_n) \text{ form a triangle}\}$.

3.3 A general hypergraph container statement with fingerprints

Lemma 3.3. (Hypergraph Container Lemma) For all $c \geqslant 1$ and $k \in \mathbb{N}$, there exists $\delta \sim 1/ck^4$ s.t the following holds. Let $\mathcal{H}$ be a k-uniform hypergraph on n-vtxs and $\tau > 0$ satisfying

$$\Delta_l(\mathcal{H}) \leqslant c\tau^{l-1}\frac{e(\mathcal{H})}{v(\mathcal{H})} = c\tau^{l-1}d$$

then there exists families $\mathcal{S} \subset \mathcal{I}(\mathcal{H})$ and $\mathcal{C} \subset P(V(\mathcal{H}))$ and a function $f : \mathcal{S} \to \mathcal{C}$ such that

(a) For every $S \in \mathcal{S}$, $|S| \leqslant \tau n$.

(b) For every $C \in \mathcal{C}$, $|C| \leqslant (1-\delta)n$.

(c) For every $I \in \mathcal{I}(\mathcal{H})$, there exists $S(I) \in \mathcal{S}$ s.t

$$S(I) \subset I \subset C(I) = f(S(I)).$$

With its full generality, we can prove the container theorem for K_3 encoding graphs.

Theorem 3.4. (Containers for triangle free graphs) For every $\varepsilon > 0$, there exists a family of fingerprints $\mathcal{S}(n,\varepsilon)$ and $\mathcal{G}(n,\varepsilon)$ of subgraphs of K_n satisfying the following properties:

(a) $|\mathcal{G}| \leqslant \exp(C(\varepsilon)n^{3/2}\log n)$.

(b) For every $G \in \mathcal{G}$, $k_3(G) \leqslant \varepsilon n^3$.

(c) For every triangle free graph H on n vtx, there exists $G \in \mathcal{G}$ with $H \subset G$.

Proof. We will iteratively apply our lemma [3.3] until we are left with our desired family. What degree properties can we say about $\mathcal{H} = \mathcal{H}_{K_3}(K_n)$? It's simple to note that $e(\mathcal{H}) = \binom{n}{3}$, $d = \Delta_1(\mathcal{H}) = n-2$ and $\Delta_2(\mathcal{H}) \leqslant 1$. So, the first time, we can apply the lemma with $\tau = 1/\sqrt{n}$.

Let $\mathcal{C} = \{E(K_n)\}$, while there is an element $E(G)$ in $\mathcal{C}$ with more than εn^3 triangles, we apply the container lemma on G, and substitute G by its containers in $\mathcal{C}$. We know the maximum degree condition is the same, and because $k_3(G) \geqslant \varepsilon n^3$, we know that the avg. degree in our container lemma is at least εn. So we can always repeat this process with $c = 1$ and $\tau = 1/\varepsilon\sqrt{n}$.

How many steps can we do? Well after t iterations, each container has size at most $(1-\delta)^t\binom{n}{2}$ yielding a maximum of $(1-\delta)^t n^3$ triangles. So we must have that $(1-\delta)^t \geqslant \varepsilon$, and therefore $t \leqslant \log(1/\varepsilon)/\delta$. So how many containers do we end up with? Something like

$$\binom{n^2}{n^{3/2}}^t = n^{O(n^{3/2})} = \exp(O(n^{3/2}\log n)).$$

We can even bound the sizes of the fingerprints by $t\tau\binom{n}{2} = Cn^{3/2}$. □

3.4 Some cool applications!

3.4.1 Counting triangle-free graphs

We will try to prove the following theorem:

Theorem 3.5. Almost all triangle free graphs with n vtxs and $m \gg n^{3/2}$ edges are $o(m)$ close to being bipartite.

To prove it we will need an estability result (basically Erdös-Simonitivs) that states

Theorem 3.6. If G is t-far from being bipartite, then

$$k_3(G) \geqslant \frac{n}{6}(e(G) + t - \frac{n^2}{4})$$

Proof. (of 3.5) Let $X(n, m, \alpha)$ be the number of triangle free graphs with m edges which are αm-far from being bipartite. We will upperbound X counting by counting inside the containers. So suppose we applied [3.4] with $\varepsilon > 0$, which yielded $\mathcal{G}$. We have

$$X(n, m, \alpha) \leqslant \sum_{G \in \mathcal{G}} \#\{K_3 \not\subset H \subset G,\ e(H) = m,\ H\ \alpha m\text{-far from being bip.}\}.$$

Now fix some parameter $\delta > 7\varepsilon$ we choose later, we know from the stability result that either G is δn^2 close to being bipartite or

$$e(G) \leqslant \frac{n^2}{4} + 6\varepsilon n^2 - \delta n^2 \leqslant \frac{n^2}{4} - \varepsilon n^2.$$

So let's separate our counting depending on the container and bound the previous sum

$$X(n, m, \alpha) \leqslant \sum_{\substack{G \in \mathcal{G} \\ e(G) \leqslant (1/4 - \varepsilon) n^2}} \binom{e(G)}{m} + \sum_{\substack{G \in \mathcal{G} \\ G\ \delta-\text{close to bip.}}} \binom{\delta n^2}{\alpha m} \binom{n^2/4}{(1-\alpha) m}$$

where the first inequality follows from taking all m subgraphs of G when G has few edges and the second follows from choosing the αm edges that make H far from being bipartite. A bit of manipulation shows

$$\begin{aligned} X(n, m, \alpha) &\leqslant \sum_{G \in \mathcal{G}} (1 - 4\varepsilon)^m \binom{n^2/4}{m} + \binom{\delta n^2}{\alpha m} \binom{n^2/4}{m} \binom{\frac{n^2}{4}}{4}^{-\alpha m} \\ &\leqslant n^{C n^{3/2}} e^{-cm} \binom{n^2/4}{m} \to o(1) \binom{n^2/4}{m} \end{aligned}$$

when $m \gg n^{3/2} \log n$.

What if we want to remove the $\log n$? Then we do the same thing but we use our fingerprints, we sum over them and remember that each S must be cointained in H. Taking that into consideration the $\log n$ vanishes.

$$X(n, m, \alpha) \leqslant \sum_{|S|=0}^{C n^{3/2}} \binom{n^2/2}{|S|} \left(\binom{(1/4 - \varepsilon) n^2}{m - |S|} + \binom{n^2/4}{(1 - \alpha/2) m - |S|} \right)$$

and after some more careful computation we win. $\square$

3.4.2 When does Gnp arrows a triangle?

The following is a very surprising theorem (and beautiful one as well).

Theorem 3.7. Let fix $r \in \mathbb{N}$ and let $p \gg n^{-1/2}$, then $G(n,p) \to_r K_3$ w.h.p.

As always, we need some kind of saturation result. We will actually show that $\mathbb{P}(G(n,p) \not\to K_3) \to 0$. So what does this event really mean? If $G(n,p) \not\to_r K_3$, then there exists a partition (or coloring) of the edges of $G(n,p) = H_1 \cup H_2 \cdots \cup H_r$, such that each H_i is K_3-free. Now an easy double counting yields the following lemma:

Lemma 3.8. Let G be a graph and χ an r-coloring of its edges. Then if

$$e(G) \geqslant (1-\delta)\binom{n}{2}$$

then χ contains δn^3 monochromatic triangles.

And we are ready to apply our containers.

Proof. (of 3.7) We upperbound again on our containers, let C_i be the containers for H_i. We know $E(K_n) = F \cup C_1 \cdots \cup C_r$ where $k_3(C_i) \leqslant \varepsilon n^3$ for each i and F is the complement of the union. By the previous lemma, it must be that $e(F) \geqslant \varepsilon r\binom{n}{2}$, and this means that, if $G(n,p) \subset C_1 \cup \cdots \cup C_r$, it is avoiding a positive density set of edges. We find that (maximizing by the last term)

$$\begin{aligned}
\mathbb{P}(G(n,p) \not\to K_3) &\leqslant \sum_{S_1 \cup \cdots \cup S_r} \mathbb{P}((S_1 \cup \dots S_r) \subset G(n,p) \subset (C_1 \cup \cdots \cup C_r)) \\
&\leqslant \sum_{t=0}^{Crn^{3/2}} \binom{n^2/2}{t} p^t (1-p)^{\delta n^2} \leqslant e^{-p\delta n^2} \left(\frac{epn^2}{Crn^{3/2}}\right)^{Crn^{3/2}} \\
&= \left(\frac{e^{-\delta p\sqrt{n}} p\sqrt{n}}{D}\right)^{Crn^{3/2}}
\end{aligned}$$

and it is clear that if $p\sqrt{n} \to \infty$, then the probability is tends to 0.

□

4 Random Graph Theory

The following is a very nice theorem by Béla!

Theorem 4.1. [Béla 87]

$$\chi(G(n,1/2)) = (1/2 + o(1))\frac{n}{\log_2(n)} \tag{3}$$

The lowerbound follows easily from the fact that $\alpha(G(n,1/2)) \leqslant 2\log_2(n)$ whp and that for any n-vtx graph G, $\chi(G) \geqslant n/\alpha(G)$.

The upperbound is a very nice application of Janson's [B.1]. We will just grab independent sets of size $(2-\varepsilon)\log_2(n)$ while we can!.

Lemma 4.2. For $\varepsilon > 0$, with very high probability, every set of vertices of $G(n, 1/2)$ with size $m \geqslant n/(\log n)^2$, has an independent set of size $(2-\varepsilon)\log_2 n$.

Proof. We will do an union bound over m sets. For that, set $k = (2-\varepsilon)$ and fix a specific set M of size m, note that

$$\mathbb{P}(\alpha(G(n, 1/2)[M]) < k) = \mathbb{P}(\alpha(G(m, 1/2) < k)).$$

Now, what is $\mathbb{P}(\alpha(G(m, 1/2) < k))$? Let $S \in \binom{[m]}{k}$ and denote $B_S = \{S \text{ is indep.}\}$. It is clear now that,

$$\mathbb{P}(\alpha(G(m, 1/2)) < k) = \mathbb{P}\left(\bigcap_S B_S^c\right)$$

and we are in the hypothesis of Janson. Note that

$$\mu = \sum_{S \in \binom{[m]}{k}} \mathbb{P}(S \text{ is indep.}) = \binom{m}{k}\left(\frac{1}{2}\right)^{\binom{k}{2}} \geqslant \left(\frac{m}{k}2^{-k/2}\right)^k \gg (n^\varepsilon)^{\log n}$$

and

$$\Delta = \sum_{\substack{S,T \in \binom{[m]}{k} \\ S \cap T > 2}} \mathbb{P}(S \cup T \text{ indep.}) = \sum_{l=2}^{k-1} \binom{m}{k}\binom{k}{l}\binom{m-k}{k-l}\left(\frac{1}{2}\right)^{2\binom{k}{2}-\binom{l}{2}}.$$

Now it's a bit ugly to see but this thing here is convex and the sum is majorized by the last two elements. We get

$$\begin{aligned}\Delta &\simeq \binom{m}{k}\binom{k}{2}\binom{m-k}{k-2}\left(\frac{1}{2}\right)^{2\binom{k}{2}-1} + \binom{m}{k}k(m-k)\left(\frac{1}{2}\right)^{\binom{k}{2}} \\ &\simeq 2\binom{m}{k}^2 \frac{k^4}{m^2}2^{-2\binom{k}{2}} + km\mu = \frac{2k^4}{m^2}\mu^2 + km\mu \leqslant \frac{3k^4}{n^2}\mu^2.\end{aligned}$$

By the second part of Janson, we get that

$$\mathbb{P}\left(\bigcap_S B_S^c\right) \leqslant \exp(-cn^2/k^4) \lll 2^{-n}$$

So we can easily do a union bound, over all sets. □

Proof of Theorem 4.1. Just use the previous lemma and remove independent sets untill you have less than $n/(\log n)^2$ vertices left. At that point, do a trivial coloring :). □

5 Some tangential topics

5.1 Discrepancy

This might become a section if we see more of it next semester. Discrepancy calculates how much polarization we must have in the edges of our graph. From now on, let $\mathcal{H}$ be a hypergraph.

Definition 5.1. Given $\mathcal{H}$ and $c : V(\mathcal{H}) \to \{-1, 1\}$, we define

$$\operatorname{disc}(\mathcal{H}, c) = \max_{S \in E(\mathcal{H})} \left| \sum_{u \in S} c(u) \right|.$$

We also define

$$\operatorname{disc}(\mathcal{H}) = \min_{c} \operatorname{disc}(\mathcal{H}, c).$$

Theorem 5.2. (Beck-Fiala) Let $\mathcal{H}$ be a hypergraph with $\Delta(\mathcal{H}) \leqslant d$, then $\operatorname{disc}(\mathcal{H}) \leqslant 2d - 1$.

A (Sub)Exponential Tailbounds

I just took these from my notes in random matrices.

Lemma A.1. (Hoeffding's Lemma) Let X be a scalar r.v taking values in $[a, b]$. Then for any $t > 0$

$$\mathbb{E}e^{tX} \leqslant e^{t\mathbb{E}X}(1 + O(t^2\sigma^2 \exp(O(t(b-a))))). \tag{4}$$

In particular

$$\mathbb{E}e^{tX} \leqslant e^{t\mathbb{E}X} \exp(O(t^2(b-a)^2)). \tag{5}$$

Proof. Normalize X as to $\mathbb{E}(X) = 0$ and $b - a = 1$. And consider the Taylor expansion of e^{tX},

$$e^{tX} = 1 + tX + O(t^2X^2 \exp(O(t)))$$

taking expectation yields the first result. The second one follows from assymptotics of the exponential function. □

Theorem A.2. (McDiamird's Theorem) Let $(R_i)_{i=1}^n$ be a tuple of ranges and $X = (X_i)_{i=1}^n$ a random vector taking values in their product. Let $F : \prod_{i=1}^n R_i \to \mathbb{R}$ be a coordinate-wise bounded function (this is a strong Lipchitz condition). That is changing any coordinate x_i may change at most c_i the total value. Then

$$\mathbb{P}(|F(X) - \mathbb{E}F(X)| > \lambda\sigma) \leqslant C \exp(-c\lambda^2)$$

for some $C, c > 0$ and $\sigma^2 = \sum_{i=1}^n c_i^2$.

Proof. By simmetry, we may only consider $\mathbb{P}(F(X) - \mathbb{E}F(X) > \lambda\sigma)$. We turn to Hoeffding's Lemma [A.1], consider

$$\mathbb{E}\exp(tF(X)) = \mathbb{E}\big[\mathbb{E}[\exp(tF(X))|X_1, \ldots X_{n-1}]\big],$$

we should try to understand $\mathbb{E}[\exp(tF(X))|X_1, \ldots X_{n-1}]$. We have that

$$\mathbb{E}[\exp(tF(X))|X_1, \ldots X_{n-1}] = \mathbb{E}[\exp(t(F(X) - \mathbb{E}(F))) \exp(t\mathbb{E}F(X))|X_1, \ldots X_{n-1}]$$

applying (5) to the first product, (here we are conditioning on $X_1, \ldots X_{n_1}$)

$$\mathbb{E}[\exp(t(F(X) - \mathbb{E}_{X_n}F(X))) \exp(t\mathbb{E}F(X))|-] \leqslant \exp(O(t^2c_n^2))\mathbb{E}[\exp(t\mathbb{E}F(X))|-]$$

and we can iteratively apply the theorem to the new coordinate-wise bounded function

$$G(X_1, \dots X_{n-1}) = \mathbb{E}[F(X)|X_1, \dots X_{n-1}].$$

Notice G has the same coefficients as F, because

$$\begin{aligned}|G(y, x_2 \dots, x_{n-1}) - G(z, x_2 \dots, x_{n-1})| &= \left| \int_{R_n} F(y, \dots, x_n) dx_n - \int_{R_n} F(z, \dots, x_n) dx_n \right| \\ &= \left| \int_{R_n} F(y, \dots, x_n) - F(z, \dots, x_n) dx_n \right| < c_1,\end{aligned}$$

and here we can take any other coordinate instead of the first. □

Corollary A.3. (Hoeffding's inequality) Let $(X_i)_{i=1}^n$ be real r.v's taking value in bounded intervals $[a_i, b_i]$. Then, if $S = \sum_{i \leqslant n} X_i$, we have

$$\mathbb{P}(|S - \mathbb{E}S| > \lambda\sigma) < C \exp(-c\lambda^2)$$

for some $C, c > 0$, where $\sigma^2 = \sum_{i \leqslant n} (b_i - a_i)^2$.

Proof. Apply [A.2] with $F(X_1, \dots X_n) = X_1 + \dots + X_n$. □

Theorem A.4. (Chernoff) Let $(X_i)_{i=1}^n$ be independent scalar random variables with $|X_i| < K$ a.s, with mean μ_i, and variance σ_i^2. Then for any $\lambda > 0$, one has

$$\mathbb{P}(|S_n - \mu| \geqslant \lambda\sigma) \leqslant C \exp(-c \min(\lambda^2, \frac{\lambda\sigma}{K}))$$

for some absolute $C, c > 0$, where $S_n = \sum_{i=1}^n X_i$, $\mu = \sum_{i=1}^n \mu_i$ and $\sigma^2 = \sum_{i=1}^n \sigma_i^2$.

Proof. The proof is again the use of the exponential moment with lemma [A.1]. Suppose we normalize each variable so that $K \leqslant 1$ and $\mu_i = 0$, then $\mu = 0$ and for any $0 \leqslant t \leqslant 1$,

$$\mathbb{E}(\exp(tS_n)) = \prod_{i=1}^n \mathbb{E}(\exp(tX_i)) = \prod_{i=1}^n (1 + O(t^2\sigma_i^2 \exp(O(t)))).$$

using the usual $1 + x \leqslant e^x$ ineq, we find

$$\mathbb{E}(\exp(tS_n)) \leqslant \exp(t\mathbb{E}(S_n)) \exp(O(t^2) \exp(O(t))\sigma^2) = \exp(O(t^2\sigma^2)),$$

now we use Markov and optimize on t following the restraint on $0 \leqslant t \leqslant 1$. (Suppose the hidden constant is D) we have that

$$\mathbb{P}(S_n \geqslant \lambda\sigma) \leqslant \exp(Dt^2\sigma^2 - t\lambda\sigma)$$

which yields the result after optimizing t following the constraints. The other inequality is equivalent.

□

Corollary A.5. (Tailbounds for the Binomial distribution) Let $B(n,p)$ be the binomial distribution. We have that

$$\mathbb{P}(|B(n,p) - np| \geqslant \delta np) \leqslant C\exp(-c\delta^2 np)$$

for some absolute C, c.

Proof. Note that

$$\mathbb{P}(|B(n,p) - np| \geqslant \lambda\sqrt{np}) \leqslant \mathbb{P}(|B(n,p) - np| \geqslant \lambda\sqrt{np(1-p)}) \leqslant C\exp(-c\lambda^2)$$

by A.4. Applying it with $\lambda = \delta\sqrt{np}$ yields the result. $\square$

Theorem A.6. (Azuma's Inequality) Let $X_1, X_2, \dots X_n$ be a sequence scalar r.v's with $|X_i| \leqslant 1$ almost surely. Assume also that we have the martingale difference property

$$\mathbb{E}(X_i | X_1, \dots, X_{i-1}) = 0$$

almost surely for all $i = 1, \dots, n$. Then for any $\lambda > 0$, the sum $S_n := X_1 + \cdots + X_n$ satisfies

$$\mathbb{P}(|S_n| \geqslant \lambda\sqrt{n}) \leqslant C\exp(-c\lambda^2)$$

for some absolute $C, c > 0$.

Proof. We prove this by calculating exponential moments and using Hoeffding's Lemma again. Note that $S_n = S_{n-1} + X_n$ so that

$$\mathbb{E}(e^{tS_n}) = \mathbb{E}(e^{tS_{n-1}} e^{tX_n}).$$

By taking the expectation of conditioning on the first $n-1$ values, we get that

$$\mathbb{E}(e^{tS_n}) = \mathbb{E}\mathbb{E}(e^{tS_{n-1}} e^{X_n} | X_1, \dots, X_{n-1}) = \mathbb{E}[e^{tS_{n-1}} \mathbb{E}(e^{tX_n} | X_1, \dots, X_{n-1})]$$

Now, by A.1, we have that $\mathbb{E}(e^{tX_n} | X_1, \dots, X_{n-1}) \leqslant e^{O(t^2)}$, so by iterating we get the result. $\square$

Theorem A.7 (Subexponential Bernstein inequality)**.** Let $X_1, \dots, X_n$ be independent, mean 0, subexponential r.v's. Then, for every $t \geqslant 0$, we have

$$\mathbb{P}\left\{\left|\sum_{i=1}^n X_i\right| \geqslant t\right\} \leqslant 2\exp\left(-c\min\left(\frac{t^2}{\sum_{j=1}^n ||X_j||_{\psi_1}^2}, \frac{t}{\max_j ||X_j||_{\psi_1}}\right)\right)$$

where c is an absolute constant.

Proof. The proof is again only using the exponential Markov method, but as in the proof of Chernoff, we have some limiting on λ as the variables are only subexponential. $\square$

B Poisson Paradigm

Theorem B.1 (Janson Inequalities)**.** Let $A_1, A_2 \dots A_t \subset [N]$ be a family of sets. Consider the events $B_j = \{A_j \subset [N]_p\}$. Then

$$\mathbb{P}\left(\bigcap_{j=1}^{t} B_j^c\right) \leqslant \begin{cases} \exp(-\mu + \Delta/2) & \\ \exp(-\mu^2/2\Delta) & \text{if } \mu \leqslant \Delta \end{cases} \tag{6}$$

where

$$\mu = \sum_{i=1}^{t} \mathbb{P}(B_j)$$

and

$$\Delta = \sum_{i \sim j} \mathbb{P}(B_i \cap B_j)$$

(here the sum being over ordered pairs).

Remark B.2. We also have the trivial lowerbound

$$\mathbb{P}(\bigcap_{j=1}^{t} B_j^c) \geqslant \prod_{j=1}^{t} \mathbb{P}(B_j^c).$$